Notes: Alfaro, Antras, Chor & Conconi (JPE, 2019)

Internalizing Global Value Chains: A Firm-Level Analysis

Contents

1 Big picture

- **Question.** Which stages of a global value chain should a firm own, and which should it leave to independent suppliers?
- **Core setting.** Production is sequential. Upstream stages make inputs that feed into downstream stages, and each stage requires relationship-specific investment.
- **Main friction.** Contracts over input compatibility and relationship-specific investments are incomplete. Ownership reallocates ex post bargaining power but does not make all contracts enforceable.
- **Main theoretical insight.** The location of integrated stages depends on whether inputs behave as *sequential complements* or *sequential substitutes*. This distinction is governed by the elasticity of final-good demand relative to the elasticity of substitution across stages.
- **Complements case.** When final demand is elastic, upstream investments raise the marginal value of downstream investments. Firms outsource upstream stages to preserve strong supplier incentives and integrate relatively downstream stages.
- **Substitutes case.** When final demand is inelastic or stages are highly substitutable, upstream and downstream investments crowd each other out. Firms integrate relatively upstream stages and outsource downstream stages.
- **Empirical innovation.** The paper combines Dun & Bradstreet WorldBase firm-activity data with input-output tables to infer where integrated and nonintegrated inputs sit along the value chain.
- **Main empirical result.** Firms in high demand-elasticity industries integrate relatively less-upstream stages; firms in low elasticity industries integrate relatively more-upstream stages.
- **Contractibility result.** The contractibility of stages upstream of an input affects integration differently depending on demand elasticity, consistent with the property-rights mechanism.
- **Productivity result.** More productive firms integrate more inputs, and the productivity effect interacts with stage location in the way predicted by the model.
- **Economic takeaway.** Global value-chain ownership is not simply a make-or-buy decision for a single input. It is an ordered, system-level decision in which the optimal boundary of the firm depends on the position of each stage, the elasticity of demand, contractibility, and firm productivity.

2 Incremental contribution

- **From theory to firm-level measurement.** Earlier work on sequential production and firm boundaries was mostly theoretical. This paper shows how standard input-output tables can be combined with global firm-establishment data to measure ownership choices along production chains.
- **Bilateral upstreamness.** Rather than using industry-level upstreamness relative to final demand, the paper constructs input-output-pair upstreamness: how far input i sits upstream in the production of output j.
- **Integrated versus nonintegrated stages.** The paper creates firm-level measures of the average upstreamness of integrated inputs and compares them with the upstreamness of inputs that are relevant but not owned.
- **A richer property-rights model.** The model generalizes Antras and Chor (2013) by allowing input asymmetries in productivity, cost, contractibility, and integration feasibility.
- **Cross-firm and within-firm tests.** The empirical strategy tests the theory both across firms and within firms across the many inputs needed to produce the parent firm's output.
- **Contractibility up to an input.** The paper introduces a cumulative upstream-contractibility measure that maps closely to the model's expression for the optimal bargaining power of the firm.

- **Sparse integration insight.** The model explains why intrafirm trade flows can be small even when ownership choices are driven by contractual incompleteness: integrated stages may be sparse and separated by outsourced stages.
- **Contribution to global value chain research.** The paper links the international fragmentation of production to the property-rights theory of the firm, providing systematic global evidence on where firms internalize stages.

3 Data and measurement

3.1 WorldBase firm and establishment data

- The empirical analysis uses Dun & Bradstreet WorldBase, which reports establishments, ownership links, primary activities, and up to six activities per establishment.
- The main sample contains more than 300,000 manufacturing parent firms in 116 countries.
- Parent firms are firms whose primary SIC code is in manufacturing and that have at least 20 employees.
- Ownership links in WorldBase identify which activities are internal to a firm and which are not.
- The paper distinguishes single-establishment firms, domestic multi-establishment firms, and multinationals, and checks robustness across subsamples.

Interpretation: WorldBase gives the ownership side of the analysis. It does not track shipments between plants, but it tells which production activities are controlled by the same firm.

3.2 Input-output construction of relevant inputs

- For each parent output industry j, the U.S. input-output table identifies inputs i used directly and indirectly to produce j.
- The direct requirements coefficient dr_{ij} is the dollar value of input i used directly to produce one dollar of output j.
- The total requirements coefficient tr_{ij} is the total direct and indirect value of input i needed to produce one dollar of j.
- Integrated inputs are activities inside the firm's ownership boundary. Nonintegrated inputs are relevant inputs from the input-output table that are not observed within the firm.

Interpretation: The input-output table supplies the production-technology side: which inputs are plausibly relevant to each output and how important they are.

3.3 Upstreamness of input-output pairs

The paper defines the upstreamness of input i in the production of output j as a weighted average of how many stages before final production input i enters:

$$Upstreamness_{ij} = \frac{dr_{ij} + 2\sum_k dr_{ik}dr_{kj} + 3\sum_k\sum_l dr_{ik}dr_{kl}dr_{lj} + \cdots}{dr_{ij} + \sum_k dr_{ik}dr_{kj} + \sum_k\sum_l dr_{ik}dr_{kl}dr_{lj} + \cdots}.$$

- $Upstreamness_{ij} = 1$ if input i only enters directly into output j.
- Larger values mean that a greater share of input i's use occurs further upstream.
- This measure is bilateral: the same input can have different upstreamness values depending on the output industry.

Interpretation: This is the main stage-position variable. It converts the abstract continuum of stages in the theory into an empirical measure for each input-output pair.

3.4 Ratio-upstreamness

$$\textit{Log Ratio-Upstreamness}_{jpc} = \log\left(\frac{\text{average upstreamness of integrated inputs}}{\text{average upstreamness of nonintegrated inputs}}\right).$$

- The ratio is computed at the parent-output-industry by parent-country level.
- Values above zero mean integrated inputs are more upstream than nonintegrated inputs.
- Values below zero mean integrated inputs are more downstream than nonintegrated inputs.

Interpretation: This dependent variable tests the cross-firm prediction that demand elasticity shifts the relative upstreamness of stages inside firm boundaries.

3.5 Demand elasticity, contractibility, and productivity

- Demand elasticities are taken from Broda and Weinstein product-level elasticities and mapped to industries.
- Higher demand elasticity moves the model toward the sequential complements case.
- Input contractibility uses Nunn-style measures based on the share of inputs traded on organized exchanges or reference-priced markets.
- Upstream-Contractibility and Contractibility-up-to-i cumulate contractibility along the production chain.
- Firm productivity is proxied mainly by sales per employee.

Interpretation: These variables map the model's primitives into observed industry and firm characteristics.

4 Models and empirical specifications

4.1 Sequential production and surplus division

- A final-good producer organizes a continuum of stages indexed by $m \in [0, 1]$.
- Suppliers undertake relationship-specific investments that affect compatibility and the value of later stages.
- Contracts are incomplete; after each stage is completed, the firm and supplier bargain over the incremental surplus.
- Integration gives the firm greater bargaining power, denoted β_V, while outsourcing gives the firm lower bargaining power, denoted β_O with $\beta_O < \beta_V$.

Interpretation: Integration is valuable because it reallocates surplus toward the firm, but it weakens supplier incentives. The optimal ownership boundary balances these forces stage by stage.

4.2 Complements versus substitutes

The key comparison is between the demand elasticity parameter ρ and the production-stage substitutability parameter α.

- If $\rho > \alpha$, stages are sequential complements.
- If $\rho < \alpha$, stages are sequential substitutes.

Interpretation: This comparison determines whether the firm should use high-powered incentives upstream or downstream.

4.3 Proposition 1: baseline firm-boundary prediction

- In the complements case, there is a cutoff m_c^* such that stages below the cutoff are outsourced and stages above the cutoff are integrated.
- In the substitutes case, there is a cutoff m_s^* such that stages below the cutoff are integrated and stages above the cutoff are outsourced.

Interpretation: This is the central theoretical result: demand conditions determine whether integration is relatively upstream or downstream.

4.4 Proposition 2: contractibility along the chain

- Higher contractibility of upstream inputs reduces the need to use ownership to fix upstream underinvestment.
- In the complements case, this lowers the range of outsourced upstream stages and therefore increases integration around the relevant margin.
- In the substitutes case, it reduces the range of integrated upstream stages.

Interpretation: Contractibility does not simply increase or decrease vertical integration. Its effect depends on where contractible stages lie and on whether stages are complements or substitutes.

4.5 Proposition 3: productivity and integration

- With fixed costs of integration, more productive firms integrate more stages.
- In the complements case, higher productivity shifts the cutoff upstream, expanding integrated downstream stages.
- In the substitutes case, higher productivity shifts the cutoff downstream, expanding integrated upstream stages.

Interpretation: Productivity increases the payoff from solving contractual distortions, so high-productivity firms internalize more of the value chain.

4.6 Cross-firm specification

$$\textit{Log Ratio-Upstreamness}_{jpc} = \gamma_0 + \sum_q \gamma_q 1\{\textit{Elasticity}_j \in q\} + \Gamma' X_{jpc} + \eta_c + \varepsilon_{jpc}.$$

Interpretation: The coefficients on high elasticity quintiles test whether integrated inputs become less upstream relative to nonintegrated inputs when the final-good demand elasticity is higher.

4.7 Within-firm specification

$$Integration_{ijp} = \gamma_0 + \sum_q \gamma_q 1\{Elasticity_j \in q\} \times Upstreamness_{ij} + \Gamma' X_{ij} + \delta_p + \varepsilon_{ijp}.$$

Interpretation: Parent-firm fixed effects compare inputs within the same firm. This removes time-invariant parent-level confounds and asks whether the firm chooses more upstream versus downstream inputs differently depending on demand elasticity.

4.8 Contractibility-up-to-i specification

$$Integration_{ijp} = \gamma_0 + \sum_q \gamma_q 1\{Elasticity_j \in q\} \times Contractibility\text{-}up\text{-}to\text{-}i_{ij} + \Gamma' X_{ij} + \delta_p + \varepsilon_{ijp}.$$

Interpretation: This is the closest empirical counterpart to the model's cumulative upstream contractibility term.

5 Identification and empirical design

5.1 Core identification logic

- The paper does not use a randomized shock. It tests whether cross-sectional and within-firm patterns line up with a tightly structured model.
- The key variation is across final-good industries in demand elasticity and across input-output pairs in upstreamness and contractibility.
- Parent fixed effects in within-firm regressions absorb all firm-level unobservables that are common across a firm's inputs.
- Input-industry fixed effects in some specifications absorb general differences in how likely particular inputs are to be integrated.

Interpretation: The empirical strategy is strongest when the same firm is compared across many potential inputs.

5.2 Why input-output data are useful

- WorldBase alone shows activities inside firms but not the entire set of relevant outsourced inputs.
- Input-output tables identify both direct and indirect inputs used in each output industry.
- Combining these sources creates a counterfactual set of potential inputs that could be integrated but are not.

Interpretation: The methodology turns firm-activity data into value-chain data.

5.3 Main threats

- Demand elasticity may be measured with error when product-level elasticities are mapped to industries.
- U.S. input-output tables may not perfectly capture production technologies in all countries.
- Ownership may reflect intangible-input transfer, managerial coordination, tax planning, or regulation, not only property-rights incentives.
- The data measure common ownership, not physical intrafirm shipments.

Interpretation: These concerns are why the paper emphasizes robust patterns across specifications rather than a single causal coefficient.

5.4 Why the evidence is persuasive

- The signs differ exactly as predicted between high- and low-demand-elasticity industries.
- The same logic appears in cross-firm averages, within-firm input-level regressions, contractibility interactions, and productivity interactions.
- The model also explains why intrafirm trade can be sparse even when ownership choices reflect contractual frictions.

Interpretation: The paper is persuasive because multiple independent empirical patterns map to distinct model predictions.

6 Section-by-section study guide

- **Introduction.** Explains why global production fragmentation requires a theory of where firms draw their boundaries along value chains.
- **Model.** Builds a sequential property-rights model and derives the upstream-versus-downstream integration predictions.
- **Data.** Combines WorldBase firm activities with input-output tables to define integrated inputs, outsourced inputs, and upstreamness.
- **Empirics.** Tests cross-firm and within-firm predictions using demand elasticity, contractibility, and productivity.
- **Conclusion.** Interprets the findings as evidence that contractual frictions shape global value-chain ownership.

7 Tables and figures

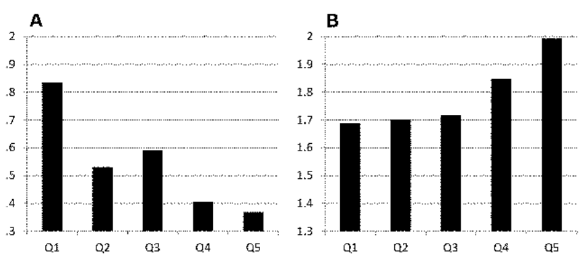
7.1 Figure 1: Average upstreamness of production stages + Figure 2: Timing of events

Read:

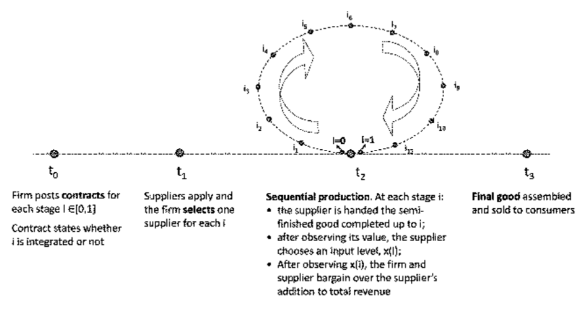
- Figure 1 plots the average upstreamness of integrated and nonintegrated stages across demand elasticity quintiles.
- Panel A shows integrated stages become less upstream as demand elasticity rises.
- Panel B shows nonintegrated stages become more upstream as demand elasticity rises.
- Figure 2 summarizes the timing of the model: contracts are posted, suppliers are selected, production proceeds sequentially, and bargaining happens after each stage's contribution is observed.

Detailed interpretation: Figure 1 is the simplest visual version of the paper's central empirical prediction. When demand is elastic, the firm wants strong incentives upstream and therefore outsources upstream stages, leaving integration relatively downstream. Figure 2 clarifies why upstream choices affect downstream incentives: production unfolds in order, so early-stage investments condition the value of later stages.

Economic message: Demand conditions shape not only whether firms integrate, but where in the value chain integration occurs.



(a) Figure 1: Average upstreamness of production stages



(b) Figure 2: Timing of events

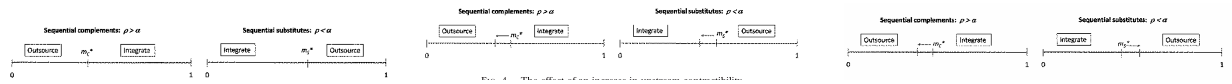
7.2 Figure 3: Firm boundary choices + Figure 4: Effect of upstream contractibility + Figure 5: Productivity effect

Read:

- Figure 3 displays the cutoff structure of ownership in complements and substitutes cases.
- Figure 4 shows how higher upstream contractibility shifts the cutoff in opposite ways depending on whether stages are complements or substitutes.
- Figure 5 shows how higher final-good productivity expands integration in both cases, but in different parts of the value chain.

Detailed interpretation: These three theoretical figures are the visual core of the model. Figure 3 gives the baseline ownership pattern. Figure 4 adds the insight that contractibility upstream changes the need for integration. Figure 5 adds the idea that more productive firms have more to gain from internalizing contractual distortions.

Economic message: The figures convert the abstract property-rights model into a small set of empirical sign predictions: demand elasticity, contractibility, and productivity should interact with stage position.



(a) Figure 3: Firm boundary choices (b) Figure 4: Upstream contractibility (c) Figure 5: Productivity effect

7.3 Table 1: Upstreamness of Integrated versus Nonintegrated Inputs

Read:

- The dependent variable is log ratio-upstreamness: integrated inputs' upstreamness relative to nonintegrated inputs' upstreamness.
- Coefficients on high elasticity quintiles are negative and often statistically significant.
- This means firms in high-elasticity industries integrate relatively more downstream inputs.
- The result survives the addition of industry controls, firm controls, country dummies, and alternative elasticity definitions.

Detailed interpretation: Table 1 is the first formal empirical test of the cross-firm prediction. Moving from low to high demand elasticity moves the model from the substitutes case toward the complements case. The negative coefficients in high quintiles show that integration shifts downstream relative to the set of relevant but nonintegrated inputs.

Economic message: The table supports the central idea that the elasticity of final demand changes where firms draw boundaries along the chain.

TABLE 1
UPSTREAMNESS OF INTEGRATED VERSUS NONINTEGRATED INPUTS

	Dependent Variable: Log Ratio-Upstreamness$_{jpc}$					
	(1)	(2)	(3)	(4)	(5)	(6)
Ind.(Quintile 2 Elas$_j$)	−.0209	−.0290	−.0278	−.0590	−.0802*	.0634
	[.0345]	[.0319]	[.0314]	[.0447]	[.0474]	[.0550]
Ind.(Quintile 3 Elas$_j$)	−.0742**	−.0802**	−.0782**	−.0569	−.0982**	−.0379*
	[.0336]	[.0316]	[.0309]	[.0454]	[.0429]	[.0224]
Ind.(Quintile 4 Elas$_j$)	−.0480	−.0893***	−.0881***	−.1068**	−.1685***	−.0942***
	[.0365]	[.0337]	[.0331]	[.0459]	[.0457]	[.0259]
Ind.(Quintile 5 Elas$_j$)	−.0588	−.0955***	−.0947***	−.1156***	−.1849***	−.1026***
	[.0377]	[.0325]	[.0318]	[.0420]	[.0459]	[.0317]
Log (Skilled Emp./Workers)$_j$		.0080	.0069	.0073	−.0290	−.0215
		[.0238]	[.0239]	[.0290]	[.0379]	[.0386]
Log (Equip. Capital/Workers)$_j$		.1127***	.1112***	.0731***	.0768***	.0949***
		[.0195]	[.0192]	[.0183]	[.0205]	[.0257]
Log (Plant Capital/Workers)$_j$		−.0331	−.0325	−.0087	−.0240	−.0316
		[.0210]	[.0207]	[.0228]	[.0276]	[.0290]
Log (Materials/Workers)$_j$		−.0311	−.0322	−.0397*	−.0099	−.0190
		[.0222]	[.0222]	[.0237]	[.0290]	[.0317]
R&D Intensity$_j$		.0053	.0044	.0113	.0048	.0017
		[.0058]	[.0057]	[.0070]	[.0086]	[.0103]
Year Started$_p$			.0001	.0001*	.0002**	.0003***
			[.0001]	[.0001]	[.0001]	[.0001]
Multinational$_p$			.0105**	.0125**	.0192**	.0304***
			[.0048]	[.0060]	[.0079]	[.0085]
Log (Total Employment)$_p$			−.0003	.0004	.0005	−.0005
			[.0016]	[.0017]	[.0019]	[.0019]
Log (Total USD Sales)$_p$			.0003	−.0004	−.0003	−.0001
			[.0008]	[.0009]	[.0011]	[.0012]
Elasticity based on	All goods	All goods	All goods	BEC cons. & cap. goods	BEC cons. goods	BEC cons. & α proxy
Parent country dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	316,977	316,977	286,072	206,490	144,107	144,107
No. of industries	459	459	459	305	219	219
R^2	.0449	.1504	.1580	.1770	.2333	.2268

NOTE.—The sample comprises all firms with primary SIC in manufacturing and at least 20 employees in the 2004/5 vintage of D&B WorldBase. The dependent variable is the log ratio-upstreamness measure described in Sec. III. Quintile dummies are used to distinguish firms with primary SIC output that are in high- vs. low-demand elasticity industries. Cols. 1–3 use a measure based on all available HS10 elasticities from Broda and Weinstein (2006); col. 4 restricts this construction to HS codes classified as consumption or capital goods in the UN BEC; col. 5 further restricts this to consumption goods; col. 6 uses the consumption-goods-only demand elasticity minus a proxy for α to distinguish between the complements and substitutes cases. All columns include parent country fixed effects. Cols. 3–6 also include indicator variables for whether the reported employment and sales data, respectively, are estimated/missing/from the low end of a range, as opposed to being from actual data (coefficients not reported). Standard errors are clustered by parent primary SIC industry.
* Significant at the 10 percent level.
** Significant at the 5 percent level.
*** Significant at the 1 percent level.

(a) Table 1, part 1 (b) Table 1, part 2

7.4 Table 2: Effect of Upstream Contractibility + Table 3: Productivity and Vertical Integration

Read:

- Table 2 interacts upstream contractibility with elasticity quintiles.
- The high-elasticity interaction is positive and significant, showing that upstream contractibility offsets the tendency to outsource upstream inputs in the complements case.
- Table 3 shows more productive firms integrate more inputs and shift the relative location of integration in the predicted direction.

- The productivity effect is especially visible in the number of integrated inputs and in ratio-upstreamness outcomes.

Detailed interpretation: Table 2 tests a more subtle prediction than Table 1: contractibility has different implications depending on the demand regime. Table 3 then shows that productivity increases the scale of integration, consistent with fixed costs of integration and higher gains from solving incentive problems.

Economic message: Contractibility and productivity do not merely explain more or less integration; they explain the location and extent of integration along the value chain.

TABLE 2
EFFECT OF UPSTREAM CONTRACTIBILITY

	DEPENDENT VARIABLE: LOG RATIO-UPSTREAMNESS$_{jpc}$			
	(1)	(2)	(3)	(4)
Ind.(Quintile 2 Elas$_j$)	−.0350	−.0611	−.0490	.0763**
	[.0300]	[.0396]	[.0429]	[.0323]
Ind.(Quintile 3 Elas$_j$)	−.1104***	−.0566	−.0683**	−.0476**
	[.0288]	[.0405]	[.0328]	[.0223]
Ind.(Quintile 4 Elas$_j$)	−.1207***	−.1605***	−.1611***	−.1185***
	[.0304]	[.0292]	[.0277]	[.0236]
Ind.(Quintile 5 Elas$_j$)	−.1409***	−.1760***	−.1643***	−.1108***
	[.0297]	[.0306]	[.0292]	[.0260]
Upstream-Contractibility$_j$				
× Ind.(Quintile 1 Elas$_j$)	−1.5540***	−1.5492***	−1.8562***	−.8114
	[.4934]	[.4177]	[.4446]	[.5369]
× Ind.(Quintile 2 Elas$_j$)	−.9810***	−.5723	−.6886	−2.0195***
	[.3165]	[.5973]	[.7621]	[.6896]
× Ind.(Quintile 3 Elas$_j$)	.3271	−.3234	−.4171	.1796
	[.2408]	[.3742]	[.3855]	[.1727]
× Ind.(Quintile 4 Elas$_j$)	.3849	1.0662***	.6855***	.9811***
	[.2867]	[.2319]	[.2106]	[.2565]
× Ind.(Quintile 5 Elas$_j$)	.7106***	1.0530***	1.1171***	1.0419***
	[.2148]	[.2149]	[.2273]	[.2275]
p-value: Q5 at median Upst.-Cont.$_j$	[.0002]	[.0001]	[.0000]	[.0000]
Elasticity based on	All goods	BEC cons. & cap. goods	BEC cons. goods	BEC cons. & α proxy
Industry controls	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes
Parent country dummies	Yes	Yes	Yes	Yes
Observations	286,072	206,490	144,107	144,107
No. of industries	459	305	219	219
R^2	.2204	.2792	.3064	.3191

NOTE.—The sample comprises all firms with primary SIC in manufacturing and at least 20 employees in the 2004/5 vintage of D&B WorldBase. The dependent variable is the log ratio-upstreamness measure described in Sec. III. Upstream-Contractibility$_j$ is the total requirements weighted covariance between the contractibility and upstreamness of the manufacturing inputs used to produce good j. Quintile dummies are used to distinguish firms with primary SIC output in high- vs. low-demand elasticity industries. Col. 1 uses a measure based on all available HS10 elasticities from Broda and Weinstein (2006); col. 2 restricts

(a) Table 2: Effect of upstream contractibility

	DEPENDENT VARIABLE					
	Log (No. of Int. Inputs)$_{jpc}$		Log Ratio-Upstreamness$_{jpc}$			
	All Inputs (1)	All Inputs (2)	All Inputs (3)	All Inputs (4)	Ever-Int. Inputs (5)	Ever-Int. Inputs (6)
Ind.(Log(Sales/Emp)$_p$ > median)						
× Ind.(Quintile 1 Elas$_j$)	.0195***	.0123	−.0026**	−.0023**	−.0029**	−.0023**
	[.0066]	[.0081]	[.0013]	[.0010]	[.0014]	[.0010]
× Ind.(Quintile 2 Elas$_j$)	.0190	.0216***	−.0002	−.0035*	−.0001	−.0036*
	[.0117]	[.0066]	[.0018]	[.0020]	[.0018]	[.0020]
× Ind.(Quintile 3 Elas$_j$)	.0342***	.0373**	.0039	.0064**	.0041	.0065**
	[.0120]	[.0171]	[.0033]	[.0027]	[.0033]	[.0027]
× Ind.(Quintile 4 Elas$_j$)	.0334***	.0286***	.0061***	.0060***	.0061***	.0059***
	[.0095]	[.0092]	[.0014]	[.0014]	[.0014]	[.0014]
× Ind.(Quintile 5 Elas$_j$)	.0212*	.0204*	.0082***	.0078***	.0082***	.0078***
	[.0109]	[.0106]	[.0024]	[.0024]	[.0024]	[.0024]
Elasticity based on	BEC cons.	BEC cons. & α proxy	BEC cons.	BEC cons. & α proxy	BEC cons.	BEC cons. & α proxy
Firm controls	Yes	Yes	Yes	Yes	Yes	Yes
Parent country-industry dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	142,135	142,135	142,135	142,135	142,135	142,135
No. of industries	219	219	219	219	219	219
R^2	.3809	.3809	.7665	.7666	.7631	.7632

NOTE.—The sample comprises all firms with primary SIC in manufacturing and at least 20 employees in the 2004/5 vintage of D&B WorldBase. The dependent variable in cols. 1 and 2 is the log number of four-digit SIC codes integrated by the firm, while that in cols. 3–6 is the log ratio-upstreamness measure described in Sec. III. In cols. 3 and 4, the dependent variable is constructed on the basis of all inputs, while in cols. 5 and 6 it is constructed on the basis of the set of ever-integrated inputs. Quintile dummies are used to distinguish firms with primary SIC output in high- vs. low-demand elasticity industries; cols. 1, 3, and 5 use the elasticity measure based only on HS10 codes classified as consumption goods in the UN BEC, while cols. 2, 4, and 6 use the consumption-goods-only demand elasticity minus the proxy for α to distinguish between the complements and substitutes cases. All columns include parent country by parent primary SIC industry pair dummies and the full list of firm-level variables used in the specifications in table 1, cols. 3–6. Standard errors are clustered by parent primary SIC industry.
* Significant at the 10 percent level.
** Significant at the 5 percent level.
*** Significant at the 1 percent level.

(b) Table 3: Productivity and vertical integration

7.5 Table 4: Integration Decisions within Firms–The Role of Upstreamness

Read:

- The unit of observation is an input by parent-firm pair.
- Parent fixed effects compare different inputs within the same firm.
- Upstreamness has the most negative effect on integration in high-elasticity industries.
- The self-SIC coefficient is very large, confirming that firms often report activities in their own primary industry, but the upstreamness pattern remains after controlling for this.

Detailed interpretation: Table 4 is important because it reduces concern that the cross-firm results simply reflect differences across firms or countries. The within-firm design asks whether a given firm integrates some inputs and not others in a way that lines up with their value-chain position.

Economic message: The core prediction survives within-firm comparison: high-elasticity firms are less likely to integrate upstream inputs.

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TABLE 4

INTEGRATION DECISIONS WITHIN FIRMS: THE ROLE OF UPSTREAMNESS

	Dependent Variable: Integration$_{ijp}$					
	(1)	(2)	(3)	(4)	(5)	(6)
Upstreamness$_{ij}$						
× Ind.(Quintile 1 Elas$_j$)	−.0043***	.0032*	.0048***	.0054***	.0036*	−.0005
	[.0014]	[.0018]	[.0018]	[.0019]	[.0020]	[.0021]
× Ind.(Quintile 2 Elas$_j$)	−.0111***	−.0042**	−.0022	−.0030	−.0044	.0035
	[.0027]	[.0020]	[.0019]	[.0019]	[.0034]	[.0023]
× Ind.(Quintile 3 Elas$_j$)	−.0102***	−.0023	.0001	−.0002	−.0028	−.0054
	[.0017]	[.0021]	[.0022]	[.0021]	[.0027]	[.0039]
× Ind.(Quintile 4 Elas$_j$)	−.0129***	.0023	.0043	.0034	.0012	.0016
	[.0033]	[.0033]	[.0030]	[.0028]	[.0023]	[.0025]
× Ind.(Quintile 5 Elas$_j$)	−.0229***	−.0169***	−.0153***	−.0146***	−.0077**	−.0079**
	[.0047]	[.0056]	[.0055]	[.0052]	[.0034]	[.0033]
Self-SIC$_{ij}$	.9664***	.9207***	.9134***	.8823***	.8517***	.8517***
	[.0033]	[.0085]	[.0091]	[.0164]	[.0177]	[.0176]
Log (Total Requirements$_{ij}$)	.0016**	.0022***	.0034***	.0028***	.0035***	.0038***
	[.0008]	[.0008]	[.0008]	[.0008]	[.0012]	[.0012]
Upstream-Complementarity$_{ij}$		.0403***	.0367***	.0174***	.0200***	.0200***
		[.0039]	[.0038]	[.0037]	[.0037]	[.0037]
Downstream-Complementarity$_{ij}$		.0284***	.0260***	.0129**	.0171***	.0163***
		[.0065]	[.0064]	[.0052]	[.0059]	[.0057]
Diff. Log (Skilled Emp./Workers)$_{ij}$			−.0170***	−.0156***	−.0213***	−.0217***
			[.0039]	[.0037]	[.0044]	[.0045]

(a) Table 4, part 1

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Diff. Log (Plant Capital/Workers)$_{ij}$			−.0015	−.0008	.0041	.0041
			[.0023]	[.0023]	[.0026]	[.0026]
Diff. R&D Intensity$_{ij}$			−.0010	.0004	.0004	.0004
			[.0006]	[.0007]	[.0006]	[.0006]
Same-SIC2$_{ij}$				.0204***	.0166***	.0160***
				[.0041]	[.0028]	[.0028]
Same-SIC3$_{ij}$				.0457***	.0416***	.0419***
				[.0149]	[.0126]	[.0127]
p-value: Upstreamness$_{ij}$, quintile 1 − quintile 5	[.0000]	[.0003]	[.0002]	[.0001]	[.0005]	[.0161]
Elasticity based on	BEC cons.	BEC cons.	BEC cons.	BEC cons.	BEC cons.	BEC cons. & α proxy
Observations	2,648,348	2,467,486	2,467,486	2,467,486	2,467,486	2,467,486
R^2	.5376	.5398	.5407	.5440	.5646	.5646
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Input industry i fixed effects	No	No	No	No	Yes	Yes
No. of i-j pairs	8,548	7,225	7,225	7,225	7,225	7,225
No. of parent firms	46,992	41,931	41,931	41,931	41,931	41,931

NOTE.—The dependent variable is a 0-1 indicator for whether the SIC input is integrated. Each observation is a SIC input by parent firm pair, where the set of parent firms comprises those with primary SIC industry in manufacturing and employment of at least 20, which have integrated at least one manufacturing input apart from the output self-SIC. The sample is restricted to the set of the top 100 ever-integrated manufacturing inputs, as ranked by the total requirements coefficients of the SIC output industry. The quintile dummies in cols. 1–5 are based on the elasticity measure constructed using only those HS10 elasticities from Broda and Weinstein (2006) classified as consumption goods in the UN BEC; col. 6 uses the consumption-goods-only demand elasticity minus a proxy for α to distinguish between the complements and substitutes cases. All columns include parent firm fixed effects, while cols. 5 and 6 also include SIC input industry fixed effects. Standard errors are clustered by input-output industry pair.

* Significant at the 10 percent level.

** Significant at the 5 percent level.

*** Significant at the 1 percent level.

(b) Table 4, part 2

7.6 Table 5: Integration Decisions within Firms–The Role of Upstream Contractibility

Read:

- Table 5 replaces simple upstreamness interactions with interactions between elasticity quintiles and contractibility-up-to-i.
- Coefficients are largest in the highest elasticity quintile.
- The p-values at the bottom reject equality between low- and high-elasticity effects in the key specifications.
- Results are robust to input-industry fixed effects and to the elasticity-minus-α proxy.

Detailed interpretation: This is the closest empirical analogue of the model's cumulative contractibility term. The table shows that when upstream stages are more contractible, firms facing high elasticity are more willing to integrate stage i, whereas the effect is much weaker in low elasticity industries.

Economic message: The contractibility evidence strengthens the property-rights interpretation because it tests a prediction that is harder to explain with a generic vertical-integration story.

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TABLE 5

INTEGRATION DECISIONS WITHIN FIRMS: THE ROLE OF CONTRACTIBILITY

	Dependent Variable: Integration$_{ijp}$					
	(1)	(2)	(3)	(4)	(5)	(6)
Contractibility-up-to-i_{ij}						
× Ind.(Quintile 1 Elas$_j$)	.0216***	−.0017	−.0065	−.0105	.0014	.0148**
	[.0050]	[.0061]	[.0064]	[.0066]	[.0063]	[.0058]
× Ind.(Quintile 2 Elas$_j$)	.0388***	.0158*	.0097	.0120	.0232***	.0020
	[.0084]	[.0084]	[.0080]	[.0074]	[.0070]	[.0067]
× Ind.(Quintile 3 Elas$_j$)	.0356***	.0093	.0035	.0032	.0221***	.0267***
	[.0053]	[.0072]	[.0075]	[.0068]	[.0073]	[.0086]
× Ind.(Quintile 4 Elas$_j$)	.0497***	−.0036	−.0085	−.0058	.0122	.0127
	[.0119]	[.0126]	[.0120]	[.0108]	[.0084]	[.0083]
× Ind.(Quintile 5 Elas$_j$)	.0822***	.0514***	.0470***	.0445***	.0418***	.0411***
	[.0144]	[.0163]	[.0161]	[.0153]	[.0108]	[.0103]
Self-SIC$_{ij}$	.9601***	.9204***	.9140***	.8827***	.8513***	.8512***
	[.0039]	[.0083]	[.0089]	[.0163]	[.0177]	[.0178]
Log (Total Requirements$_{ij}$)	.0003	.0016	.0028***	.0023**	.0013	.0015
	[.0009]	[.0010]	[.0010]	[.0009]	[.0011]	[.0011]
Upstream-Complementarity$_{ij}$		.0393***	.0360***	.0167***	.0205***	.0207***
		[.0041]	[.0039]	[.0037]	[.0036]	[.0037]
Downstream-Complementarity$_{ij}$		.0278***	.0255***	.0124**	.0172***	.0159***
		[.0069]	[.0067]	[.0054]	[.0058]	[.0056]
Diff. Log (Skilled Emp. Workers)$_{ij}$			−.0169***	−.0154***	−.0210***	−.0209***
			[.0039]	[.0036]	[.0044]	[.0044]

(a) Table 5, part 1

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Diff. Log (Equip. Capital/Workers)$_{ij}$			−.0029	−.0034	−.0087***	−.0087***
			[.0021]	[.0021]	[.0030]	[.0030]
Diff. Log (Plant Capital/Workers)$_{ij}$			−.0015	−.0008	.0042	.0041
			[.0023]	[.0022]	[.0027]	[.0027]
Diff. R&D Intensity$_{ij}$			−.0011*	.0003	.0003	.0003
			[.0006]	[.0007]	[.0006]	[.0006]
Same-SIC2$_{ij}$				.0203***	.0160***	.0153***
				[.0040]	[.0028]	[.0028]
Same-SIC3$_{ij}$				.0461***	.0422***	.0429***
				[.0148]	[.0126]	[.0127]
p-value: Contractibility-up-to-i_{ij}, quintile 1 − quintile 5	[.0000]	[.0007]	[.0005]	[.0002]	[.0001]	[.0068]
Elasticity based on	BEC cons.	BEC cons	BEC cons.	BEC cons.	BEC cons.	BEC cons. & α proxy
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Input industry i fixed effects	No	No	No	No	Yes	Yes
Observations	2,648,348	2,467,486	2,467,486	2,467,486	2,467,486	2,467,486
No. of parent firms	46,992	41,931	41,931	41,931	41,931	41,931
No. of i-j pairs	8,548	7,225	7,225	7,225	7,225	7,225
R^2	.5383	.5397	.5406	.5438	.5647	.5647

NOTE.—The dependent variable is a 0-1 indicator for whether the SIC input is integrated. Each observation is a SIC input by parent firm pair, where the set of parent firms comprises those with primary SIC industry in manufacturing and employment of at least 20, which have integrated at least one manufacturing input apart from the output self-SIC. The sample is restricted to the set of the top 100 ever-integrated manufacturing inputs, as ranked by the total requirements coefficients of the SIC output industry. The Contractibility-up-to-i_{ij} measure is the share of the total-requirements weighted contractibility of inputs that has been accrued in production upstream of and including input i in the production of output j. The quintile dummies in cols. 1–5 are based on the elasticity measure constructed using only those HS10 elasticities from Broda and Weinstein (2006) classified as consumption goods in the UN BEC; col. 6 uses the consumption-goods-only demand elasticity minus a proxy for α to distinguish between the complements and substitutes cases. All columns include parent firm fixed effects, while cols. 5 and 6 also include SIC input industry fixed effects. Standard errors are clustered by input-output industry pair.

* Significant at the 10 percent level.

** Significant at the 5 percent level.

*** Significant at the 1 percent level.

(b) Table 5, part 2

8 Result map

- **Prediction P.1 Cross.** A firm's propensity to integrate upstream relative to downstream inputs falls with demand elasticity. Supported by Table 1.
- **Prediction P.1 Within.** Within a firm, upstreamness has a more negative effect on integration when demand elasticity is high. Supported by Table 4.

- **Prediction P.2 Cross.** Upstream contractibility interacts with demand elasticity in shaping relative upstreamness. Supported by Table 2.
- **Prediction P.2 Within.** Contractibility-up-to-i affects integration more strongly for high-elasticity industries. Supported by Table 5.
- **Productivity prediction.** Higher productivity expands integration, with different upstream/downstream implications across demand regimes. Supported by Table 3.
- **Sparse integration prediction.** Even if integration is rare and intrafirm trade is small, ownership decisions can still follow the model's logic along feasible stages.

9 Short summary

- The paper studies how firms decide which stages of a global value chain to own.
- It develops a property-rights model of sequential production with incomplete contracts.
- Integration gives the firm greater bargaining power but weakens supplier incentives.
- The location of integration depends on whether stages are sequential complements or substitutes.
- High demand elasticity pushes firms to outsource upstream and integrate downstream.
- Low demand elasticity pushes firms to integrate upstream and outsource downstream.
- The data combine WorldBase establishment activities with input-output tables.
- The main empirical measure is the upstreamness of integrated relative to nonintegrated inputs.
- Cross-firm and within-firm regressions support the model's key predictions.
- Contractibility and productivity provide additional tests that line up with the theory.
- The paper explains why intrafirm trade flows can be small even when ownership choices reflect contractual frictions.

10 Conclusion

10.1 Core conclusion

Alfaro, Antras, Chor, and Conconi show that firm boundaries along global value chains are systematically related to the position of production stages, demand elasticity, contractibility, and firm productivity. The evidence supports a property-rights view of global production in which ownership reallocates bargaining power and changes supplier incentives.

10.2 Economic intuition

A firm does not simply choose whether to vertically integrate. It chooses where to integrate in an ordered production process. If upstream suppliers' investments are especially important for downstream incentives, giving those upstream suppliers high-powered incentives through outsourcing can be optimal. If instead upstream and downstream investments substitute for each other, upstream integration can be optimal.

10.3 Incremental takeaway

The paper's major contribution is methodological and conceptual: it shows that firm-boundary theory can be tested at the level of value-chain positions by combining firm-activity data with input-output tables.

This turns a previously abstract theory of sequential ownership into measurable cross-firm and within-firm predictions.

10.4 Policy and managerial implications

The results imply that the organization of global value chains depends on more than trade costs and tax incentives. Improvements in contract enforcement, changes in demand conditions, or shifts in firm productivity can reorganize which parts of production firms internalize. For managers, the lesson is that supplier ownership should be evaluated jointly across the chain, not input by input in isolation.

10.5 Limitations

The data measure common ownership and reported activities, not all physical flows of goods. Input-output tables are U.S.-based and may not perfectly describe every country's production technology. The evidence is consistent with the property-rights mechanism, but some patterns may also be rationalized by theories based on intangible-input transfer, coordination, or transaction-cost economics.

10.6 Final takeaway

The paper provides a template for studying the internal organization of global production. It demonstrates that the boundary of the firm is a value-chain object: ownership depends on where a stage sits, how contractible surrounding stages are, how elastic final demand is, and how productive the firm is.